

I

II B
Maths II A Formulas (TS and AP)

1.) Circles (High weightage)

- Equation of tangent at (x_1, y_1)
 $T = 0 \Rightarrow xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c = 0$
- Length of tangent from $P(x_1, y_1)$
 $\sqrt{g^2 + f^2 - c}$
- pair of tangents : $S_1^2 = S \cdot T_1$
- Radical axis between circles : $S_1 - S_2 = 0$
- Common Chord : Difference of the two circle equations.
- Conditional for orthogonal circles : $2g_1g_2 + 2f_1f_2 = c_1 + c_2$

2.) Parabola

- Equation in parametric form
 $x = at^2, y = 2at$
- length of latus rectum = $4a$
- Equation of chord joining points $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$
 $t_1t_2x - a(t_1+t_2)y + 2a^2(t_1+t_2) = 0$
- Condition for tangency $c = a/m$

3.) Ellipse

- $b^2 = a^2(1 - e^2)$
- Directrix $x = \pm a/e$
- Focal radius $r_1 = a(1 - e \cos \theta)$
- Equation of tangent at (x_1, y_1)
$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$
- Equation of normal at (x_1, y_1)
$$a^2xx_1 - b^2yy_1 = a^2b^2$$

4.) Hyperbola

- $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- Eccentricity $e = \sqrt{1 + b^2/a^2}$
- Equation of tangent $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

5.) Integration & Definite Integrals

- $\int \sec x \, dx = \log |\sec x + \tan x| + C$
- $\int \operatorname{cosec} x \, dx = \log |\operatorname{cosec} x - \cot x| + C$
- $\int \sin^2 x \, dx = (x/2) - (\sin 2x)/4 + C$
- $\int \cos^2 x \, dx = (x/2) + (\sin 2x)/4 + C$
- $\int \sec^2 x \, dx = \tan x + C$
- $\int \tan x \, dx = \log |\sec x| + C$
- $\int F'(x)/F(x) \, dx = \log |F(x)| + C$
- $\int e^{ax} \sin bx \, dx = e^{ax} (a \sin bx - b \cos bx) / (a^2 + b^2) + C$
- $\int_0^a F(x) \, dx = \int_0^a F(a-x) \, dx$

6.) Complex Numbers & De Moivre's Theorem

→ $x = |z| = \sqrt{x^2 + y^2}$, $\tan \theta = y/x$

→ Polar form: $z = x(\cos \theta + i \sin \theta)$

→ De Moivre's Theorem

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

→ Cube roots of unity: $1, \omega, \omega^2$ where $\omega = (-1 + i\sqrt{3}/2)$

→ $\omega^3 = 1, 1 + \omega + \omega^2 = 0$

7.) Theory of Equations

→ $\alpha + \beta = -b/a, \alpha\beta = c/a$

→ Relation of roots & coefficients (for cubic)

$$\alpha + \beta + \gamma = -b/a, \alpha\beta + \beta\gamma + \gamma\alpha = c/a, \alpha\beta\gamma = -d/a$$

→ Transformation of roots $x = y + h$

→ Reciprocal equation condition: Coefficients symmetric or alternately opposite.

8.) Permutations & Combinations

→ $nPr = n! / (n-r)!$

→ $nCr = n! / [r!(n-r)!]$

→ $nCr = nC(n-r)$, and $nCr+1 / nCr = (n-r) / (r+1)$

→ Number of diagonals in n -sided polygon $= n(n-3)/2$

9.) Binomial Theorem

$$(x+a)^n = \sum (nC_r x^{n-r} a^r)$$

→ Middle term

→ if n odd $\rightarrow T((n+1)/2), T((n+3)/2)$

if n even $\rightarrow T(n/2+1)$, if n odd $\rightarrow T((n+1)/2), T((n+3)/2)$

→ Greatest term $= T(r+1), r = [(na)/(x+a)]$

10.) Probability

$$\rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\rightarrow P(A|B) = P(A \cap B) / P(B)$$

$\rightarrow$ Bayes Theorem

$$P(E_k|A) = [P(E_k) P(A|E_k)] / \sum [P(E_i) P(A|E_i)]$$